Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

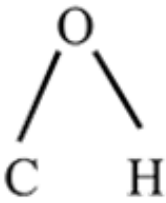
Which one of the following statements is not correct?

(A) Alcohols are weaker acids than water.

(B) Acid strength of alcohols decreases in the following $RCH_2OH > R_2CHOH > R_3COH.$

(C) Carbon-oxygen bond length in methanol, CH_3OH is shorter than that of $C-O$ bond length in phenol.

(D) The bond angle

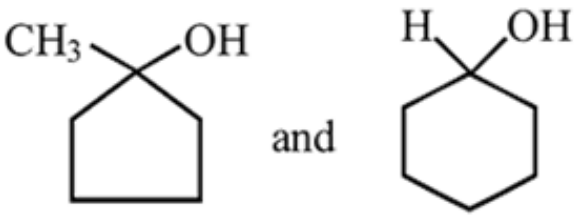


in methanol is $108.9°$.

Q.2

In the given pair of alcohol, in which pair second alcohol is more reactive than first towards hydrogen bromide?

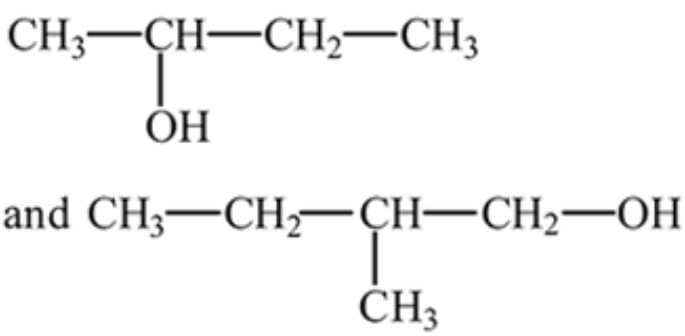
(A)



(B)



(C)

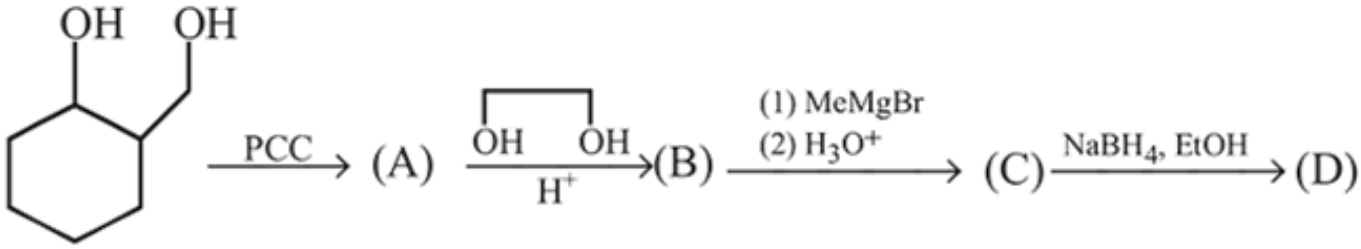


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(D) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$

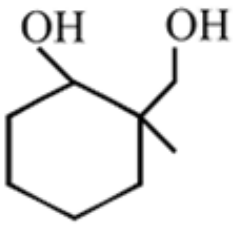
and $(CH_3)_2-\underset{\underset{OH}{|}}{C}-CH_2-CH_3$

Q.3

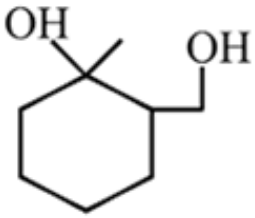


Product (D) in above reaction is:

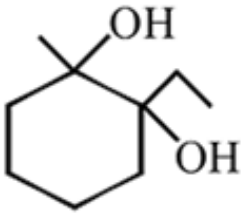
(A)



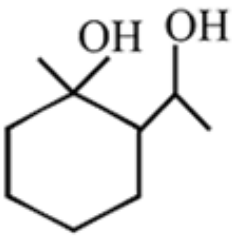
(B)



(C)



(D)



Q.4

In the Victor-Meyer's test, the colour given by 1°, 2° and 3° alcohols are respectively:

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(A) Red, colourless, blue

(B) Red, blue, colourless

(C) Colourless, red, blue

(D) Red, blue, violet

Q.5

Rate of dehydration of alcohols follows the order:

(A) $2^{\circ} > 1^{\circ} > CH_3OH > 3^{\circ}$

(B) $3^{\circ} > 2^{\circ} > 1^{\circ} > CH_3OH$

(C) $2^{\circ} > 3^{\circ} > 1^{\circ} > CH_3OH$

(D) $CH_3OH > 1^{\circ} > 2^{\circ} > 3^{\circ}$

Q.6

Amongst the following alcohols which would react fastest with conc. HCl and $ZnCl_2$?

(A) pentanol

(B) 2 -methyl butanol

(C) 2-pentanol

(D) 2-methyl butan-2-ol

Q.7

Reagent used to convert allyl alcohol to acrolein is

(A) MnO_2

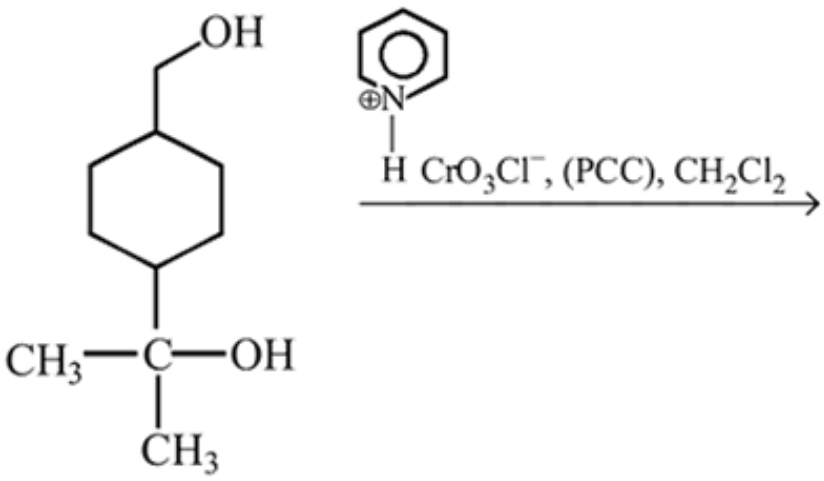
(B) H_2O_2

(C) OsO_4

(D) $KMnO_4$

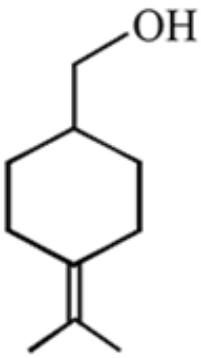
Q.8

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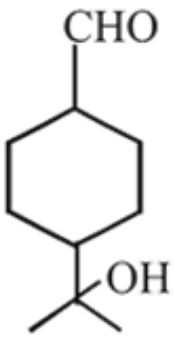


Product of the reaction is:

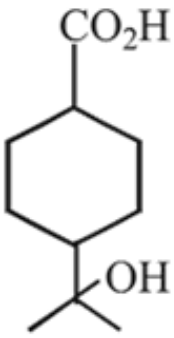
(A)



(B)

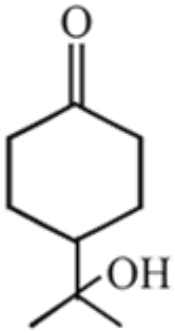


(C)



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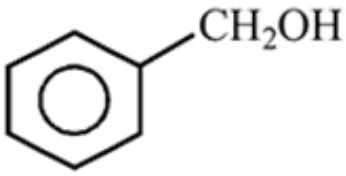
(D)



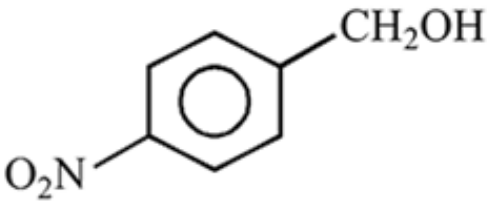
Q.9

Consider the following alcohols,

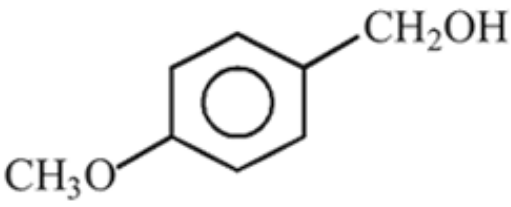
(I)



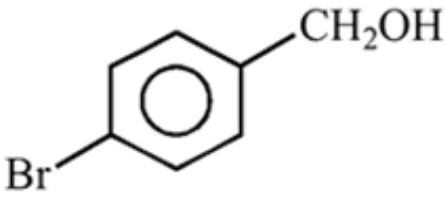
(II)



(III)



(IV)



The order of decreasing reactivities of these alcohols towards nucleophilic substitution with HBr is:

(A) III > I > IV > II

(B) III > I > II > IV

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(C) $\mathrm{I} > \mathrm{III} > \mathrm{IV} > \mathrm{II}$

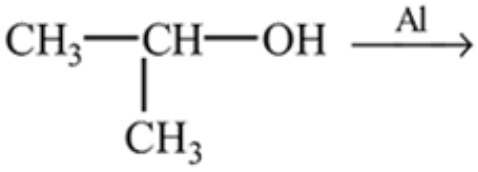
(D) $\mathrm{I} > \mathrm{III} > \mathrm{II} > \mathrm{IV}$

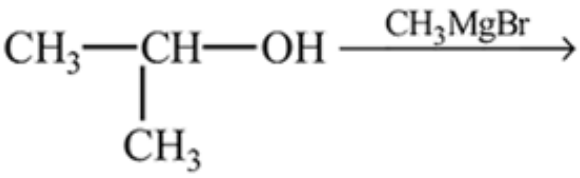
Q.10

In which of the following reactions hydrogen gas will not be evolved?

(A) $\mathrm{CH_3 - CH_2 - OH} \xrightarrow{\mathrm{Na}}$

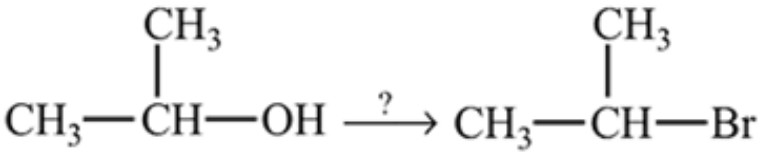
(B) $\mathrm{CH_3 - CH_2 - OH} \xrightarrow{\mathrm{K}}$

(C) 

(D) 

Q.11

Which is the best reagent to convert isopropyl alcohol to isopropyl bromide?



(A) HBr

(B) $\mathrm{SOBr_2}$

(C) $\mathrm{Br_2}$

(D) $\mathrm{CH_3MgBr}$

Q.12

On reaction with sodium, $1\ \mathrm{mol}$ of a compound X gives $1\ \mathrm{mol}$ of $\mathrm{H_2}$. Which one of the following compounds might be X?

(A) $\mathrm{CH_3CH_2CH = CH_2}$

(B) $\mathrm{CH_3COOCH_2CH_3}$

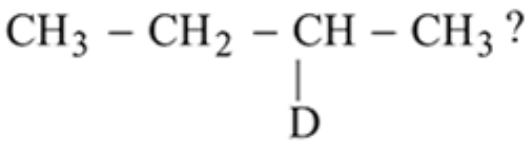
(C) $\mathrm{CH_3CH_2CH_2OH}$

(D) $\mathrm{CH_2(OH)CH_2CH_2CH_2OH}$

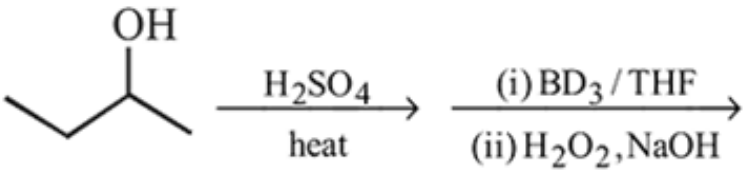
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Q.13

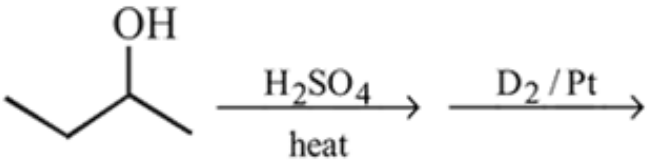
Which of the following reactions would convert 2-butanol into deuterated compound



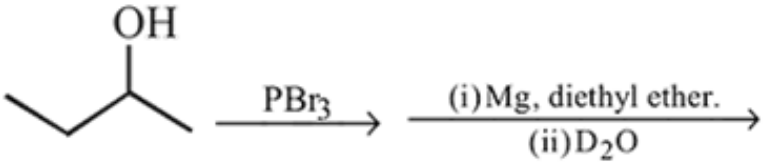
(A)



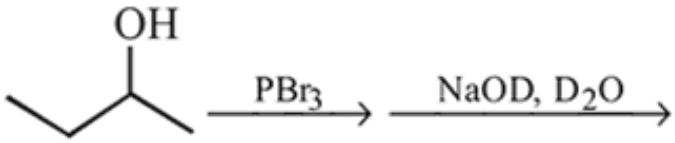
(B)



(C)



(D)



Q.14

To prepare 3 -ethylpentan-3-ol, the reagents needed are:

(A) $CH_3CH_2MgBr + CH_3COCH_2CH_3$

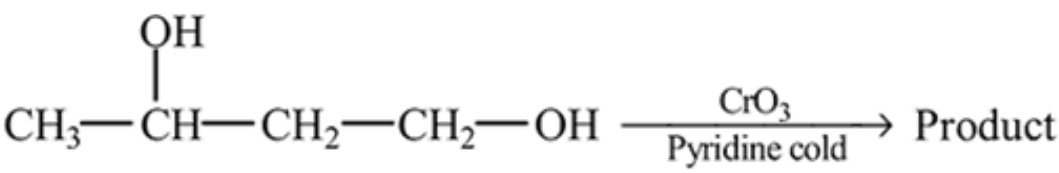
(B) $CH_3MgBr + CH_3CH_2CH_2COCH_2CH_3$

(C) $CH_3CH_2MgBr + CH_3CH_2COCH_2CH_3$

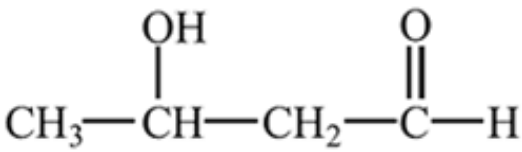
(D) $CH_3CH_2CH_2MgBr + CH_3COCH_2CH_3$

Q.15

What is the major product of the following reaction?

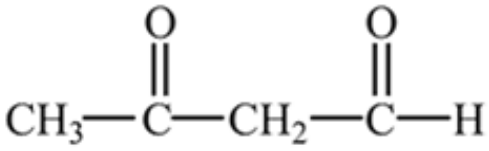


(A)

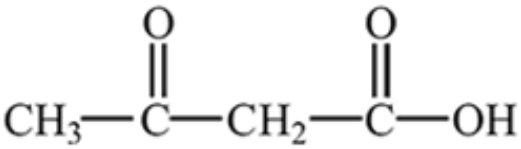


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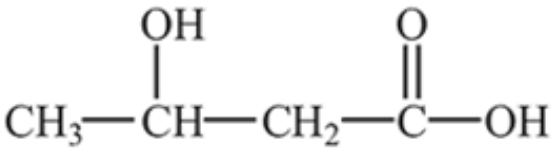
(B)



(C)



(D)



Q.16

A compound of the formula $C_4H_{10}O$ reacts with sodium and undergoes oxidation to give a carbonyl compound which does not reduce Tollen's reagent, the original compound is

(A) Diethyl ether

(B) n -Butyl alcohol

(C) Isobutyl alcohol

(D) sec-Butyl alcohol

Q.17

Identify X in the following sequence:

$$\underset{(X)}{C_3H_8O} \xrightarrow[H_2SO_4]{K_2Cr_2O_7} C_3H_6O \xrightarrow[I_2+NaOH]{\text{warm}} CHI_3$$

(A) $CH_3CH_2CH_2OH$

(B) $CH_3CH(OH)CH_3$

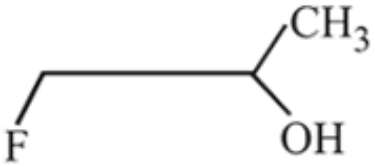
(C) $CH_3OCH_2CH_3$

(D) CH_3CH_2CHO

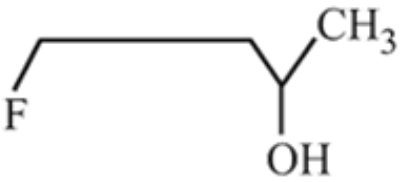
Q.18

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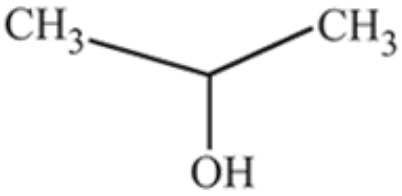
The order of reactivity of the following alcohols towards conc. HCl is



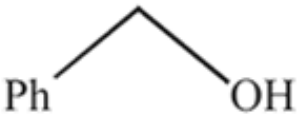
(I)



(II)



(III)



(IV)

(A) I > II > III > IV

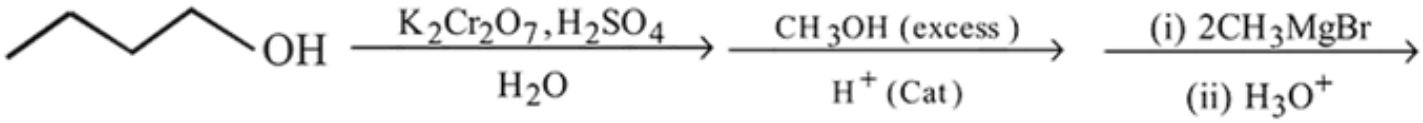
(B) I > III > II > V

(C) IV > III > II > I

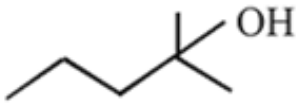
(D) IV > III > I > II

Q.19

The end product of the following sequence of reactions



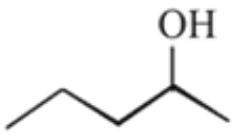
(A)



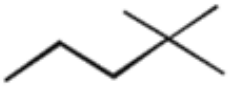
(B)



(C)



(D)



Q.20

Which of the following is not true in case of reaction with heated copper at $300^{\circ}C$?

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(A) Phenol $\longrightarrow$ Benzyl alcohol

(B) Primary alcohol $\longrightarrow$ Aldehyde

(C) Secondary alcohol $\longrightarrow$ Ketone

(D) Tertiary alcohol $\longrightarrow$ Olefin

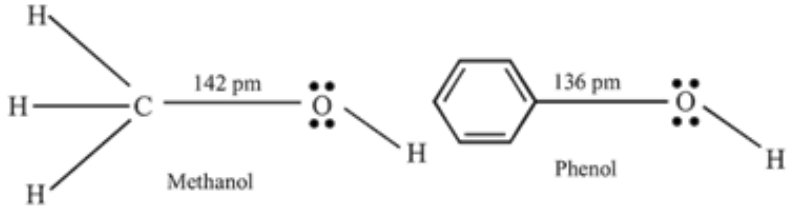
Answers & Solutions

Q.1 Answer:

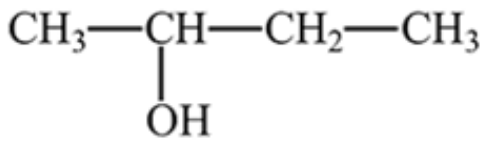
Carbon-oxygen bond length in methanol, CH_3OH is shorter than that of $C-O$ bond length in phenol.

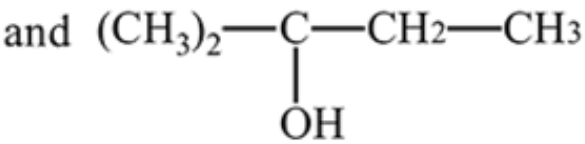
Solution:

The $C-O$ bond length in alcohols is $142pm$ and in Phenol it is $136pm$. The $C-O$ bond length in phenol is shorter than methanol it is due to the conjugation of unshared pair of electrons on oxygen with the ring, which imparts double bond character to the $C-O$ bond.

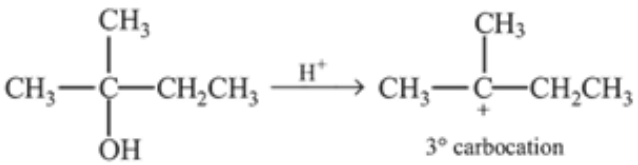


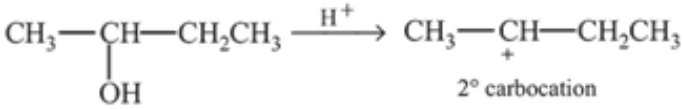
Q.2 Answer:



and 

Solution:

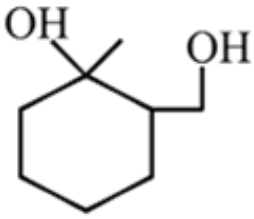




Stability order of carbocation $3^\circ > 2^\circ$

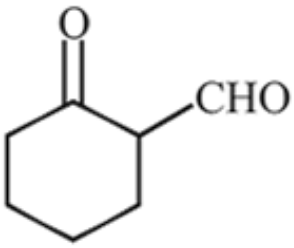
Q.3 Answer:

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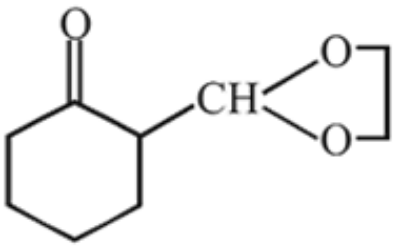


Solution:

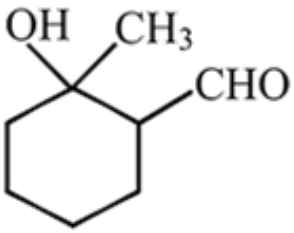
(A)



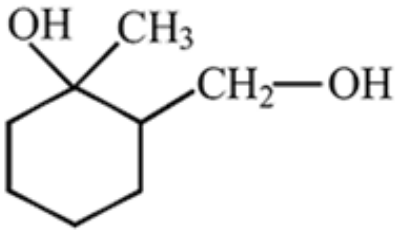
(B)



(C)



(D)



Q.4 Answer:

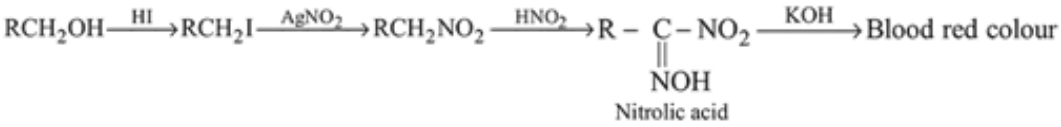
Red, blue, colourless

Solution:

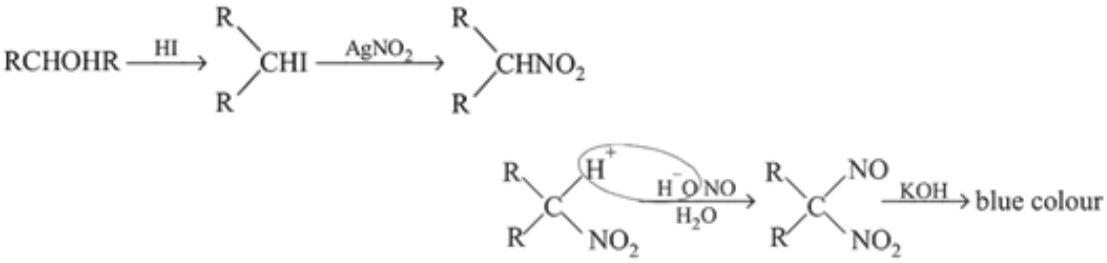
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Victor Meyer's test: The various steps involved are

(i)



(ii)



(iii)

$$R_3COH \xrightarrow{HI} R_3CI \xrightarrow{AgNO_2} R_3CNO_2 \xrightarrow{HNO_2} \text{No reaction} \xrightarrow{KOH} \text{Colourless}$$

Q.5 Answer:

$3^\circ > 2^\circ > 1^\circ > CH_3OH$

Solution:

The order of dehydration among three type of alcohols is $3^\circ > 2^\circ > 1^\circ > CH_3OH$. This behaviour is related to the relative stabilities of carbocations $(3^\circ > 2^\circ > 1^\circ)$

Q.6 Answer:

2-methyl butan-2-ol

Solution:

$ZnCl_2$ + conc. HCl is Lucas reagent. Lucas reagent reacts fastest with tertiary alcohol 3° alcohol + Lucas reagent = Immediate turbidity

Q.7 Answer:

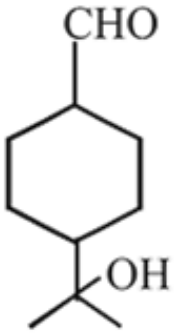
MnO_2

Solution:

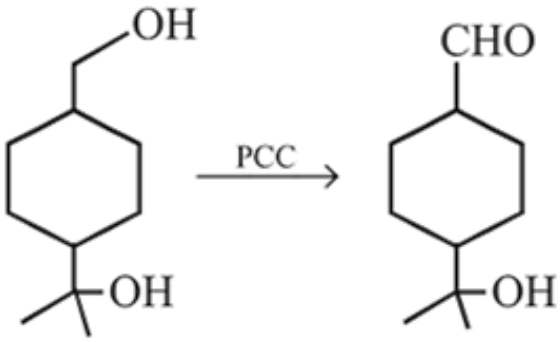
MnO_2 being a mild oxidising agent stops the oxidation of $-CH_2OH$ group at aldehyde stage.

Q.8 Answer:

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Solution:



(3° alcohol cannot be oxidized)

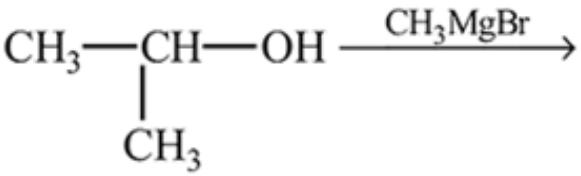
Q.9 Answer:

III > I > IV > II

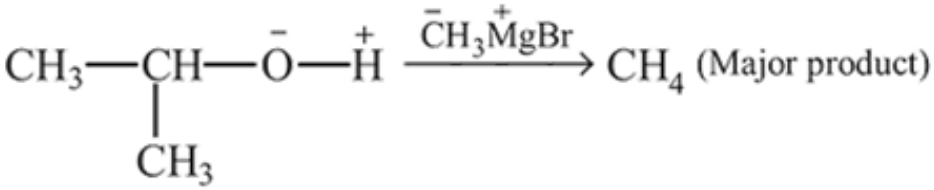
Solution:

More stable carbocation, more is the rate toward HBr (acid).

Q.10 Answer:



Solution:



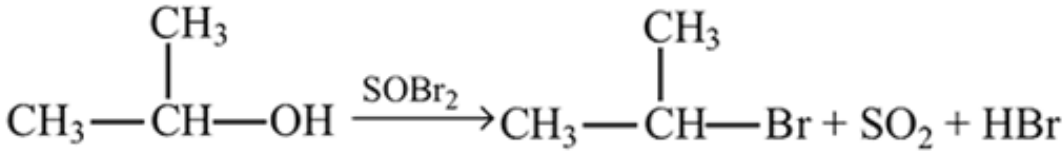
Acid-base reaction takes place

Q.11 Answer:

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$SOBr_2$

Solution:



Side products are gases.

In case of HBr elimination also favours to produce alkene.

Q.12 Answer:

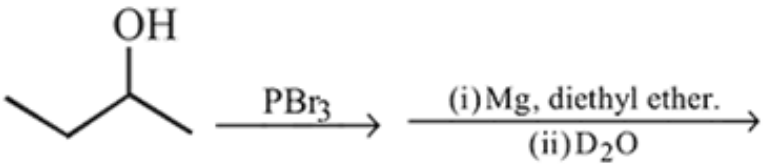
$CH_2(OH)CH_2CH_2CH_2OH$

Solution:

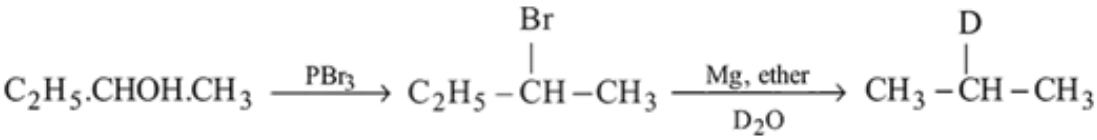
Since 1 mole of compound X reacts with Na to evolve 1 mole of H_2 gas, therefore the compound should have 2 active hydrogen atoms per mole which is possible only in option d.

$$CH_2OHCH_2CH_2CH_2OH + 2Na \longrightarrow NaOCH_2CH_2CH_2CH_2ONa + 2H$$

Q.13 Answer:



Solution:

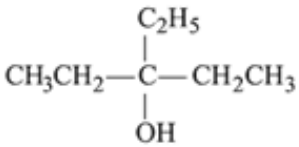


Q.14 Answer:

$CH_3CH_2MgBr + CH_3CH_2COCH_2CH_3$

Solution:

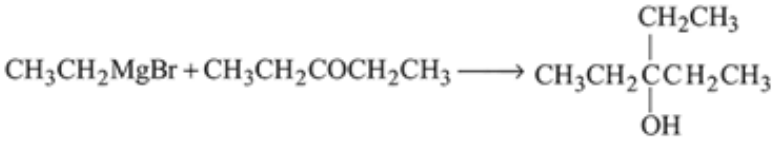
First write down the structure of the required compound, observe the nature of functional group,



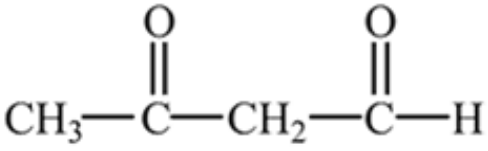
and review the memory for preparing the required functional group. Here the required compound is

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a 3° alcohol having three ethyl groups, so only combination given in option c will introduce three ethyl groups.



Q.15 Answer:

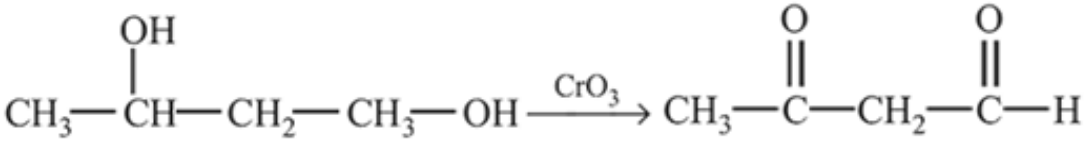


Solution:

CrO_3/ **pyridine is mild oxidising agent.**

1° **Alcohol** $\rightarrow$ **Aldehyde**

2° **Alcohol** $\rightarrow$ **Ketone ;**



Q.16 Answer:

sec-Butyl alcohol

Solution:

Since the compound $(C_4H_{10}O)$ **react with sodium, it must be alcohol (option** b, c, **or** d **). As it is oxidised to carbonyl compound which does not reduce Tollen's reagent, the carbonyl compound should be a ketone and thus** $C_4H_{10}O$ **should be a secondary alcohol, i.e. sec-butyl alcohol; other two given alcohols are** 1°**.**

Q.17 Answer:

$CH_3CH(OH)CH_3$

Solution:

Since the compound X (C_3H_6O) **gives iodoform test, it must have** $-COCH_3$ **groups hence it should be** CH_3COCH_3**. Therefore** X **should be**$CH_3CH(OH)CH_3$.

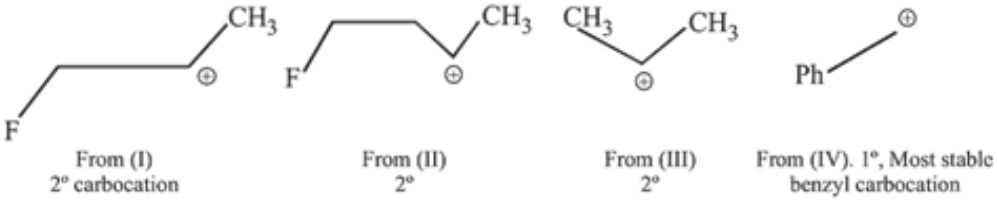
Q.18 Answer:

$IV > III > II > I$

Solution:

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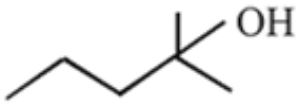
The order of reactivity depends upon the stability of the carbocations formed.



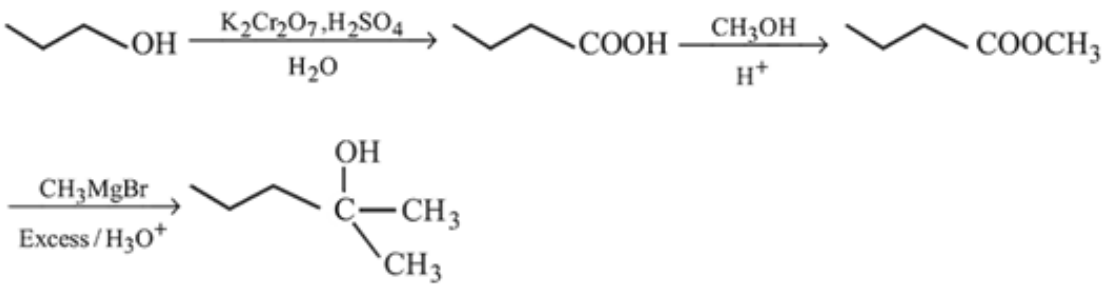
Remember that presence of electron-withdrawing group intensifies i.e., destabilises the carbocation thus (i) and (ii) are less stable than (iii). Further (i) is less stable than (ii) because-I effect is more pronounced in (i) due to less distance between F and positive charge. Thus the stability order of the four carbocations and reactivity of their parent alcohols will be

$$IV > III > II > I$$

Q.19 Answer:



Solution:



Q.20 Answer:

Phenol $\longrightarrow$ Benzyl alcohol

Solution:

Phenol does not react with heated Cu at 300°C.

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